Vineyard City 2023 Annual Drinking Water Quality Report



2023 Annual Drinking Water Quality Report

We are pleased to present you this year's Annual Drinking Water Quality Report. This report is designed to inform you about the water quality and services we deliver to you daily. Our constant goal is to provide you with a safe and dependable drinking water supply. We want you to understand how to continually improve the water treatment process and protect our water resources. We are committed to ensuring the quality of your water. Our water sources have been determined to be from surface water sources. Our water sources are Orem City and Central Utah Water.

The Drinking Water Source Protection Plan for Vineyard City is available for review. It contains information about source protection zones, potential contamination sources and management strategies to protect our drinking water. Our sources have been determined to have a medium level of susceptibility to possible contamination from septic tanks, roads, residential areas, and industrial areas. We have also developed management strategies for contamination protect our sources form contamination further. Please contact us with questions or concerns about our source protection plan.

CROSS CONNECTIONS

There are many connections to our water distribution system. When connections are properly installed and maintained, the concerns are very minimal. However, unapproved and improper piping changes or connections can adversely affect the availability and quality of the water. A cross connection may let polluted water or even chemicals mingle into the water supply system when improperly protected. This not only compromises the water quality but can also affect your health. So, what can you do? Do not make or allow improper connections at your homes. Even that unprotected garden hose in the puddle next to the driveway is a cross-connection. The unprotected lawn sprinkler system after you have fertilized or sprayed is also a cross-connection. When the cross connection is allowed at your home it will affect you and your family first. If you'd like to learn more about helping protect the quality of our water, call us for further information about how you can help.

This report shows our water quality and what it means to you, our customer.

We want our valued customers to be informed about their water utility. If you have questions about this report or your water utility, please contact the Vineyard City Public Works Water Division:

Sullivan Love at 801-376-0419 or Devan Peterson at 385-207-3893

Council meetings are also a way for residents and business owners to interface with elected officials to get involved in our community. Council meetings are held on the second and fourth Wednesday of each month at 6:00 PM, exceptions: November and December - second Wednesday only. City Council meetings are held at the Vineyard City Hall building at 125 S Main Street, Vineyard, UT. You can also contact your council members. Contact the Vineyard City Recorder at (801)226-1929, for information on how to contact them.

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Vineyard City routinely monitors constituents in our drinking water in accordance with Federal and Utah State laws. The following table shows the results of our monitoring from January 1st to December 31st, 2023. All drinking water, including bottled drinking water, may reasonably expected to contain at least small amounts of some constituents. It's important to remember that the presence of these constituents does not necessarily pose a health risk.

Please refer to page 5 for definitions

WATER QUALITY TEST RESULTS

Contaminant	Violation Y/N	Level Detected ND/Low-High	Unit Measure-ment	MCLG	MCL	Date Sampled	Likely Source of Contamination
MICROBIOLOGICAL CONTAMINANTS							
Total Coliform Bacteria	N	ND	N/A	0	Presence of coliform bacteria in 5% of monthly samples	2023	Naturally present in the environment
Fecal coliform and E.coli	N	ND	N/A	0	If a routine sample and repeat sample are total coliform positive, and one is also fecal coliform or E. coli positive	2023	Human and animal fecal waste
Turbidity for Surface Water	N	0.51	NTU	N/A	0.5 in at least 95% of the samples and must never exceed 5.0	2023	Soil Runoff (highest single measurement & the lowest monthly percentage of samples meeting the turbidity limits)
INORGANIC CONTAMINANTS							
Arsenic	N	ND-1	ppb	0	10	2023	Erosion of natural deposits; runoff from orchards; runoff from glass and electronics production wastes
Barium	N	65-87	ppb	2000	2000	2023	Discharge of drilling wastes; discharge from metal refineries; erosion of natural deposits
Carbon, Total Organic (TOC)	N	2-3	ppm	NA	TT	2023	Naturally present in the environment
Chromium	N	51	ppb	100	100	2023	Discharge from steel and pulp mills; erosion of natural deposits

Please refer to page 5 for definitions

WATER QUALITY TEST RESULTS CONTINUED

Contaminant	Violation Y/N	Level Detected ND/Low-High	Unit Measure-ment	MCLG	MCL	Date Sampled	Likely Source of Contamination
INORGANIC CONTAMINANTS CONT.							
Copper a.90% results # of sites that exceed the AL	N	a.556 b.0	ppb	1300	AL=1300	2023	Corrosion of household plumbing systems; erosion of natural deposits
Cyanide	N	ND-4	ppb	200	200	2021	Discharge from steel/metal factories; discharge from plastic and fertilizer factories
Fluoride	N	27	ppb	4000	4000	2023	Erosion of natural deposits; water additive which promotes strong teeth; discharge from fertilizer and aluminum factories
Lead a.90% results b. # of sites that exceed the AL	N	a. 5 b. 0	ppb	0	AL=15	2023	Corrosion of household plumbing systems, erosion of natural deposits
Nickel	N	17	ppb	10000	10000	2023	
Selenium	N	0.6-4	ppb	50	50	2023	Discharge from petroleum and metal refineries; erosion of natural deposits; discharge from mines
Sodium	N	18-40	ppm	None set by EPA	None set by EPA	2023	Erosion of natural deposits; discharge from refineries and factories; runoff from landfills.
Sulfate	N	8-63	ppm	1000	1000	2023	Erosion of natural deposits; discharge from refineries and factories; runoff from landfills, runoff from cropland
TDS (Total Dissolved solids)	N	112-444	ppm	2000	2000	2023	Erosion of natural deposits
DISINFECTION BY-PRODUCTS							
TTHM (Total tri-halomethanes)	N	30-31	ppb	0	80	2023	By-product of drinking water disinfection
Haloacetic Acids	N	13	ppb	0	60	2023	By-product of drinking water disinfection
RADIOACTIVE CONTAMINANTS							
Alpha emitters	N	ND-4	pCi/1	0	15	2023	Erosion of natural deposits
Radium 228	N	0.5	pCi/1	0	5	2023	Erosion of natural deposits

WATER QUALITY DATA DEFINITIONS

In the above table, you may have found terms and abbreviations you might not be familiar with. To help you better understand these terms, we've provided the following definitions:

Non-Detects (ND)
Laboratory analysis indicates that the constituent is not present

ND/Low - High
For water systems that have multiple sources of water, the Utah Division of Drinking Water has given water systems the option of listing the test results of the constituents in one table, instead of multiple tables. To accomplish this, the lowest and highest values detected in the multiple sources are recorded in the same space in the report table.

Parts per million (ppm) or Milligrams per liter (mg/l)
One part per million corresponds to one minute in two years or a single penny in $10,000.

Parts per billion (ppb) or Micrograms per liter (ug/l)
One part per billion corresponds to one minute in 2,000 years, or a single penny in $10,000,000.

Parts per trillion (ppt) or Nanograms per liter (nanograms/l)
One part per trillion corresponds to one minute in 2,000,000 years, or a single penny in $10,000,000,000.

Parts per quadrillion (ppq) or Picograms per liter (picograms/l)
One part per quadrillion corresponds to one minute in 2,000,000,000 years or one penny in $10,000,000,000,000.

Picocuries per liter (pCi/L)
Picocuries per liter is a measure of the radioactivity in water.

Millirems per year (mrem/yr)
Measure of radiation absorbed by the body.

Million Fibers per Liter (MFL)
Million fibers per liter is a measure of the presence of asbestos fibers that are longer than 10 micrometers.

Nephelometric Turbidity Unit (NTU)
Nephelometric turbidity unit is a measure of the clarity of water. Turbidity in excess of 5 NTU is just noticeable to the average person.

WATER QUALITY DATA DEFINITIONS CONTINUED

Action Level (AL)
The concentration of a contaminant which, if exceeded, triggers treatment or other requirements which a water system must follow.

Treatment Technique (TT)
A treatment technique is a required process intended to reduce the level of a contaminant in drinking water.

Maximum Contaminant Level (MCL)
The “Maximum Allowed” (MCL) is the highest level of a contaminant that is allowed in drinking water. MCLs are set as close to the MCLGs as feasible using the best available treatment technology.

Maximum Contaminant Level Goal (MCLG)
The “Goal”(MCLG) is the level of a contaminant in drinking water below which there is no known or expected risk to health.MCLGs allow for a margin of safety.

Maximum Residual Disinfectant Level (MRDL)
The highest level of a disinfectant allowed in drinking water. There is convincing evidence that addition of a disinfectant is necessary for control of microbial contaminants.

Maximum Residual Disinfectant Level Goal (MRDLG)
The level of a drinking water disinfectant below which there is no known or expected risk to health. MRDLGs do not reflect the benefits of the use of disinfectants to control microbial contaminants.

Date
Because of required sampling time frames i.e. yearly, 3 years, 4 years and 6 years, sampling dates may seem outdated.

Waivers (W)
Because some chemicals are not used or stored in areas around drinking water sources, some water systems have been given waivers that exempt them from having to take certain chemical samples, these waivers are also tied to Drinking Water Source Protection Plans.

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WHAT ARE LEAD'S HEALTH EFFECTS?

Infants and young children are typically more vulnerable to lead in drinking water than the general population. It is possible that lead levels at your home may be higher than at other homes in the community because of materials used in your home's plumbing. If you are concerned about elevated lead levels in your home's water, you may wish to have your water tested and flush your tap for 30 seconds to 2 minutes before using tap water. Additional information is available from the Safe Drinking Water Hotline (1-800-426-4791).

If present, elevated levels of lead can cause serious health problems, especially for pregnant women and young children. Lead in drinking water is primarily from materials and components associated with service lines and home plumbing. Vineyard City is responsible for providing high-quality drinking water but cannot control the variety of materials used in plumbing components. When your water has been sitting for several hours, you can minimize the potential for lead exposure by flushing your tap for 30 seconds to 2 minutes before using water for drinking or cooking. If you are concerned about lead in your water, you may wish to have your water tested. Information on lead in drinking water, testing methods, and steps you can take to minimize exposure is available from the Safe Drinking Water Hotline at (800) 426-4791 or http://www.epa.gov/safewater/lead.

100%

As you can see from the table, our system had no violations. We're proud that your drinking water meets or exceeds all Federal and State requirements. However, we have learned that some constituents have been detected through our monitoring and testing.



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UNDERSTANDING DRINKING WATER CONTAMINANTS AND HEALTH RISKS

All sources of drinking water are subject to potential contamination by naturally occurring or manmade constituents. Those constituents include microbes, organic or inorganic chemicals, or radioactive materials. All drinking water, including bottled water, may reasonably be expected to contain at least small amounts of some contaminants. The presence of contaminants does not necessarily indicate that the water poses a health risk. More information about contaminants and potential health effects can be obtained by calling the Environmental Protection Agency's (EPA) Safe Drinking Water Hotline at (800) 426-4791.

Maximum Contaminant Levels (MCLs) are set at very stringent levels. To understand the possible health effects described for many regulated constituents, a person would have to drink 2 liters of water daily at the MCL level for a lifetime to have a one-in-a-million chance of having the described health effect.

Some people may be more vulnerable to contaminants in drinking water than the general population. Immunocompromised persons such as persons with cancer undergoing chemotherapy, persons who have undergone organ transplants, people with HIV/AIDS or other immune system disorders, some elderly, and infants can be particularly at risk from infections. These people should seek advice from their healthcare providers about drinking water. EPA/CDC guidelines on appropriate means to lessen the risk of infection by cryptosporidium and other microbiological contaminants are available from the Safe Drinking Water Hotline at (800) 426-4791.

We at Vineyard City work around the clock to provide top-quality water to every tap. We ask that all our customers help us protect our water sources, the heart of our community, our way of life, and our children's future.

